

Hackathon Best Practices

Problem selection. Elicitation. Validation. Ideation. Ranking.

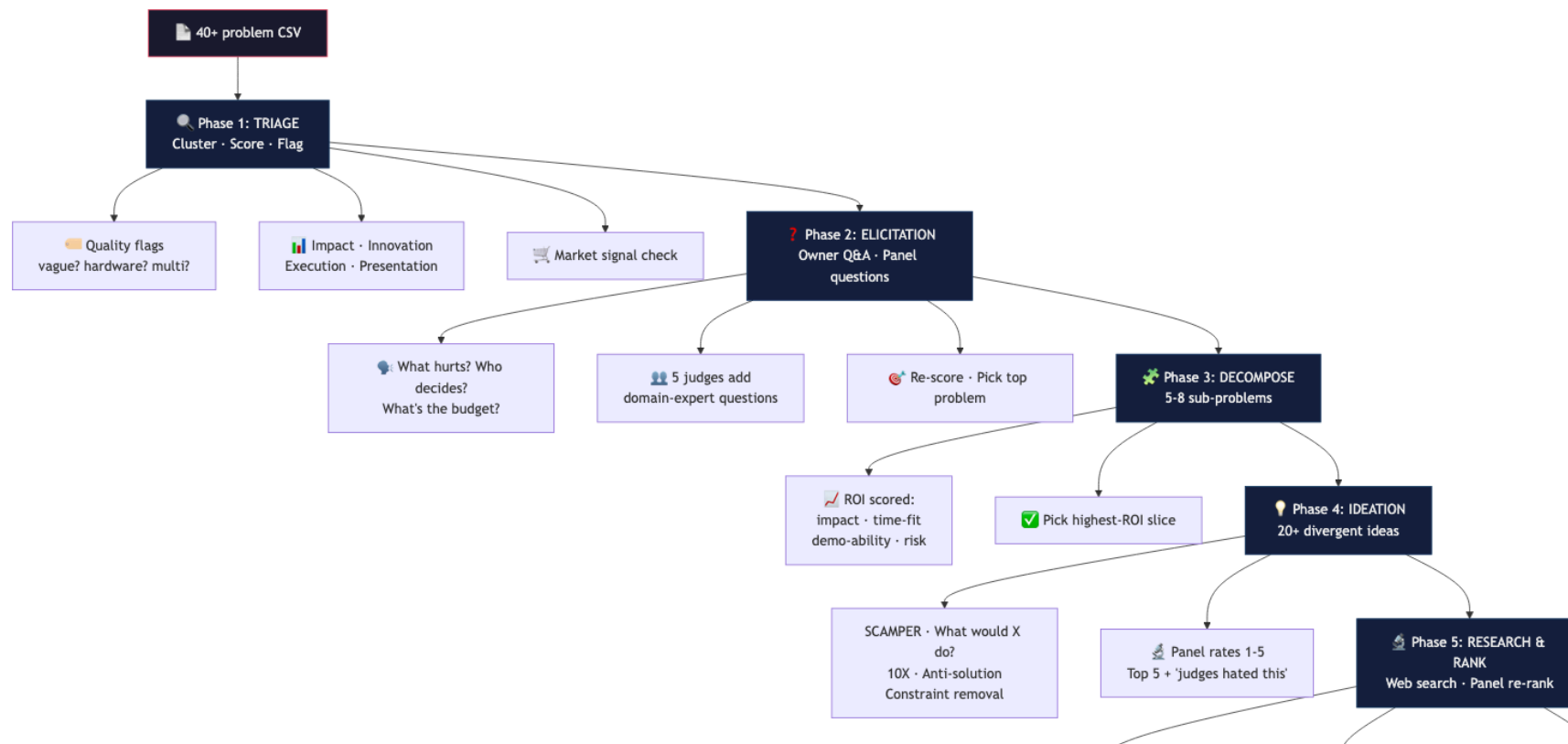
An agent that guides you through every step — with 12 domain-expert judges reviewing your work.

Built on [Marp](#)

The Goal

Most hackathon teams wing it. They pick a problem on gut feel, brainstorm 3 obvious solutions, and hope.

This agent enforces **battle-tested practices**:



Phase 1: Problem Selection

Goal: Pick the right problem — not the first one, not the easiest one.

40+ raw problems

Cluster
by theme



3 clusters
with scores

scored on:
impact, innovation,
execution, presentation

Quality
flags

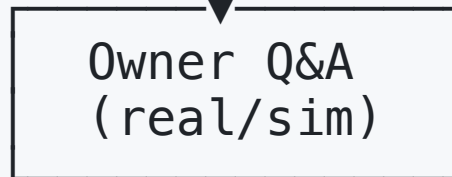
vague? multi-problem? requires-hardware?
Flag early. Kill fast.

Market-signal check per cluster: *"Are there funded competitors? What's already deployed?"*

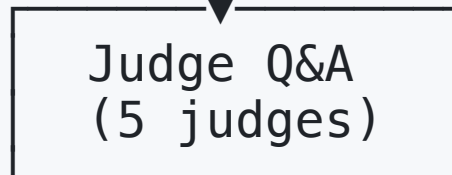
Phase 2: Problem Elicitation

Goal: Stop solving the wrong problem. Ask the owner.

Top-3 candidates



What hurts most? What did you try?
Who decides? What defines success?
What's the environment?



Each panel judge adds 2-3 hard questions
from their domain expertise

Re-score candidates with answers → User picks 1

In `owner_mode: real`, the agent asks YOU these questions. In `sim`, it role-plays a defense industry persona.

Phase 3: Sub-Problem Decomposition

Goal: Break the problem into tractable chunks. Pick the highest-ROI slice for 48 hours.

Chosen problem

5-8 sub-
problems

SP-1: Real-time feed ingestion
SP-2: Threat prioritization engine
SP-3: COA recommendation generator
SP-4: Operator-facing dashboard
SP-5: Multi-domain data fusion

ROI scored

impact · time-fit · demo-ability · dependency-risk

Panel reviews: each judge scores every sub-problem independently. Convergent picks = strong signal.

Phase 4: Divergent Ideation

Goal: Generate 20+ ideas before converging. Avoid "obvious = best."

7 techniques:

SCAMPER
(3 ideas)

What
would X
do? (3)

10X
version
(3 ideas)

Anti-
solution
(3 ideas)

Constraint
removal
(3 ideas)

Analogy
transfer
(3 ideas)

Wildcard
(2+ ideas)

Dedupe
Jaccard

Panel rates
each 1-5

Top 5 + "judges hated this"

Phase 5: Validation & Ranking

Goal: Research, don't guess. Let 5 domain experts rank, not just your team.

Top-5 ideas

Research

Panel
re-rank

Owner
picks

Web search per idea: prior art, SOTA, funded competitors, TRL estimates, regulatory pathway

Each judge re-scores with research in mind.
Weighted Borda aggregation.
Spread = genuine disagreement? (good!)

Owner (real or sim) validates the ranking.
Can override. Dissents recorded.

Phase 6: Demo & Narrative

Goal: Hackathons are won in the demo, not the deck.

Chosen solution

Thin demo

Smallest thing that demonstrates the "wow" moment

3-min
script

0:00 cold open → 0:20 problem → 0:40 live demo
→ 2:00 how it works → 2:40 closing
12+ timed beats

Q&A prep

8-12 questions from judges with prepared answers
+ risk register (demo crashes, domain gaps...)

The 12-Judge Panel

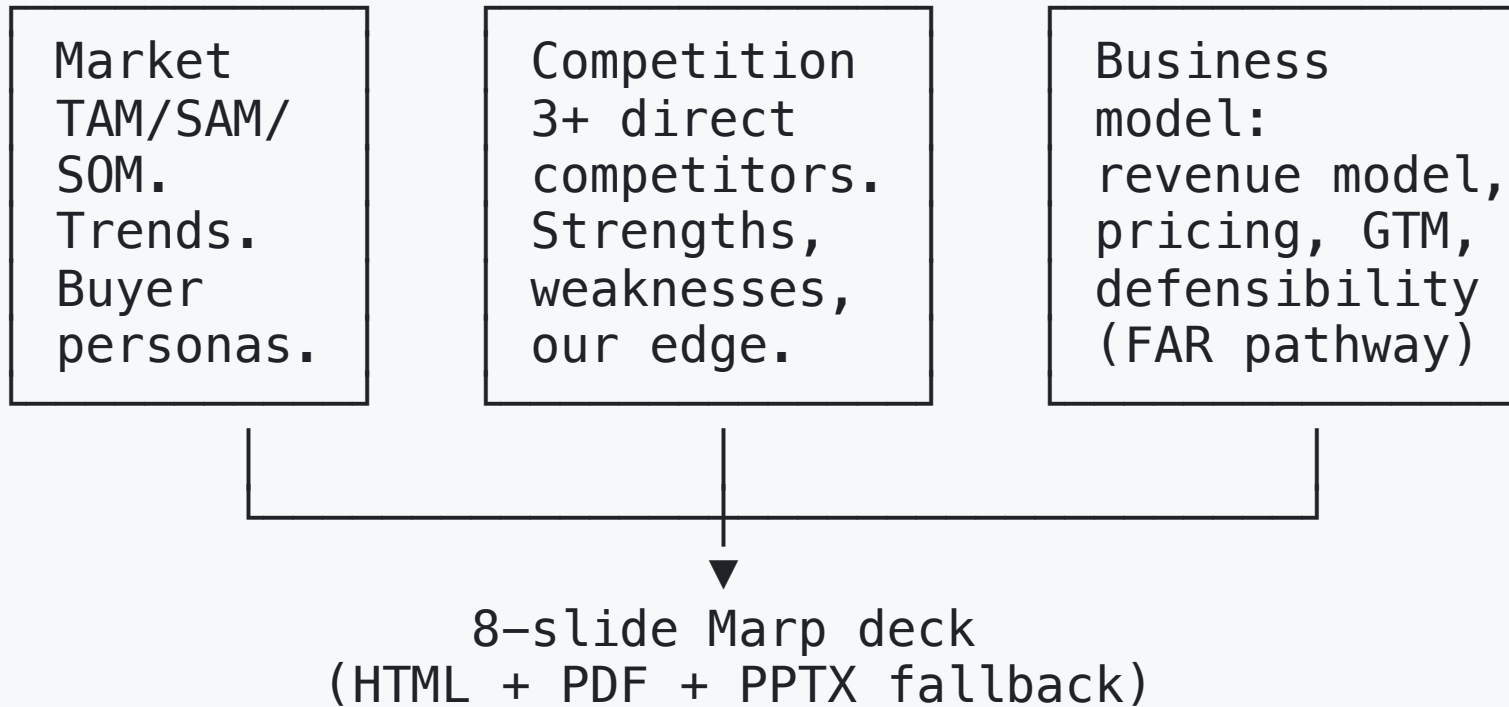
Cross-functional review at every phase. **No yes-men.**

Judge	Role	Pet peeve
Viper	F-16 pilot, 2200 hrs	"Anything requiring a PhD to operate"
Tran	EW specialist, MITRE	"Hand-waved detection thresholds"
Whitfield	Defense prime PM	"No understanding of the buyer"
Mehta	CTO, ex-Palantir (always-on)	"Buzzword soup"
Kovalenko	Ukrainian drone operator	"Delicate hardware"
Volkov	Red-team adversary	"I'd defeat this in 6 months"

Plus: procurement officer, ethics lawyer, defense VC, scaling engineer, intel analyst, UX designer.

Phase 7: Market & Deck

Goal: Show you understand why it matters, who pays, and who else is trying.



Install & Try

```
git clone <repo> && cd edth-prob-solution-deck  
python3 -m venv .venv && source .venv/bin/activate  
pip install -e ".[dev]"  
npm install -g @marp-team/marp-cli # optional, for PDF
```

One Command to Prove It Works

```
/edth-agent dry-run
```

30 seconds. Full pipeline. Zero interaction. Open `artefacts/07_deck.html` .

```
/edth-agent run      → Start a real run (interactive)
/edth-agent validate → Check all artefacts for issues
/edth-agent skip-to 5 → Generate stubs for 0-4, jump to ranking
```

Thank You

Problem → Solution. With rigour.